

Regulatory Variant Report

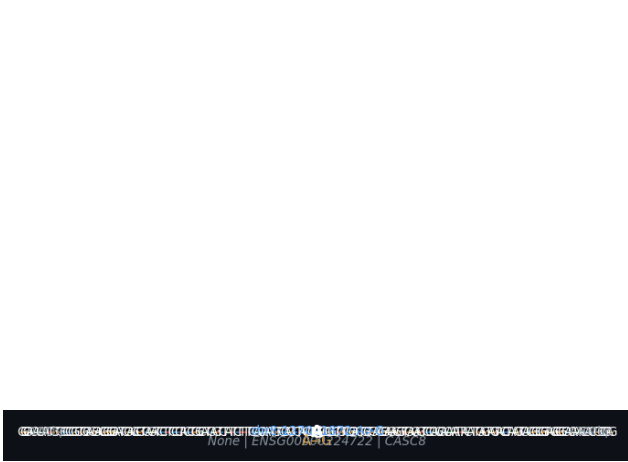
AlphaGenome + DeepSeek Agent Analysis

2026-06-03 · regvar v0.1.0

chr8:127091872:A>G

chr8:s183373024L None, ENSG00000224722, CASC8

Unranked



Scoring Summary

Assay	Gene	Biosample	Raw	Quantile
ChIP-seq histone	None	C4-2B	-0.24	-1.00
RNA-seq	ENSG00000224722	MCF-7	-0.01	-1.00
RNA-seq	CASC8	foreskin keratinocyte	0.01	1.00
RNA-seq	ENSG00000224722	HeLa-S3	-0.01	-1.00
ChIP-seq histone	None	22Rv1	-0.22	-1.00
ChIP-seq TF	None	MCF-7	-0.30	-1.00
RNA-seq	ENSG00000224722	mammary epithelial cell	-0.01	-1.00
RNA-seq	CASC8	keratinocyte	0.01	1.00
RNA-seq	CASC8	hair follicular keratinocyte	0.01	1.00
ChIP-seq histone	None	LNCaP clone FGC	-0.20	-1.00
RNA-seq	ENSG00000224722	urinary bladder	-0.01	-1.00
RNA-seq	ENSG00000224722	A549	-0.01	-1.00
ChIP-seq histone	None	VCaP	-0.24	-1.00
RNA-seq	ENSG00000224722	stomach	-0.01	-1.00
RNA-seq	ENSG00000224722	H1	-0.01	-0.99

Interpretation

chr8:127091872:A>G (rs183373024) — OR = 2.91 | 8q24 region (MYC/PCAT1)** **The most mechanistically informative hit.** This is the only variant with strong **ChIP-seq histone** effects specifically in prostate cancer lines: **C4-2B** (castration-resistant), **22Rv1** (AR-positive), **LNCaP clone FGC** (androgen-sensitive), and **VCaP** (AR-amplified, bone metastasis). Also a strong ChIP-seq TF signal in MCF-7. The location at chr8:127,091,872 places it within

the ****8q24 prostate cancer risk region**** housing MYC enhancers and the lncRNA PCAT1. Predicted to alter histone modification landscapes — likely H3K27ac or H3K4me1 at enhancers.